safe standby mode while awaiting instructions from Earth. As it ages, more and more failures are expected, both in its mechanical parts and in its computer system, although there is no sign, even now, of a serious memory deterioration, some robotic Alzheimer's disease.

This is not to say that *Voyager* is perfect. Serious mission-threatening, white-knuckle mishaps did occur. Each time, special teams of engineers—some of whom had been with the *Voyager* program since its inception—were assigned to "work" the problem. They would study the underlying science and draw upon their previous experience with the failed subsystems. They would experiment with identical *Voyager* spacecraft equipment that had never been launched, or even manufacture a large number of components of the sort that failed in order to gain some statistical understanding of the failure mode.

In April 1978, almost eight months after launch, and while the ship was approaching the asteroid belt, an omitted ground command—a human error—caused *Voyager 2*'s on-board computer to switch from the prime radio receiver to its backup. During the next ground transmission to the spacecraft, the backup receiver refused to lock onto the signal from Earth. A component called a tracking loop capacitor had failed. After seven days in which *Voyager 2* was entirely out of contact, its fault protection software suddenly commanded the backup receiver to be switched off and the prime receiver to be switched back on. Mysteriously—to this day, no one knows why—the prime receiver failed moments later. It was never heard from again. To top it off, the on-board computer now foolishly insisted on using the failed primary receiver. Through an unlucky concatenation of human and robotic error, the spacecraft was now in real jeopardy. No one could think of a way to get *Voyager 2* to revert to the backup receiver. Even if it did, the backup receiver couldn't receive the commands from Earth—because of that failed capacitor. There were many project personnel who feared that all was lost.

But after a week of obdurate unresponsiveness to all commands, instructions to switch automatically between receivers were